

Code-Layout, Readability and Reusability

Date	1 July 2026
Team ID	XXXXXX
Project Name	AI-Powered Lead Generation System
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

The AI-Powered Lead Generation System is developed using a clean, modular, and well-structured coding approach. The project separates functionality into different modules such as data preprocessing, lead prediction, lead scoring, recommendation generation, database management, and the Streamlit user interface. Meaningful variable and function names, proper indentation, comments, and exception handling improve code readability and maintainability. The project follows good software engineering practices, making the code reusable, scalable, and efficient for future enhancements.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Python code follows standard indentation and formatting guidelines (PEP 8).
2	Proper File Structure	Files and folders are logically organized	Yes	Project is organized into app.py, models, datasets, preprocessing modules, database files, and utility functions.

3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables such as customer_data, lead_score, predicted_lead, probability_score, and recommendation_list are self-explanatory.
4	Function / Method Names	Functions are descriptively named	Yes	Functions like preprocess_data(), predict_lead(), calculate_lead_score(), and generate_recommendation() clearly indicate their purpose.
5	Code Comments	Inline and block comments explain logic	Yes	Comments are included in data preprocessing, model training, prediction, and recommendation modules.
6	Modular Design	Code is split into reusable functions/modules	Yes	Separate modules are used for data processing, ML prediction, lead scoring, database handling, and Streamlit application.
7	No Redundant Code	Duplicate or unused code is removed	Partial	Some repeated data validation and processing logic can be further optimized into reusable helper functions.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
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1	Lead Prediction Model	Python, Scikit-learn, Machine Learning Algorithms	Used for predicting potential customer leads during use	High
2	Data Preprocessing Module	Python, Pandas, NumPy	Used for cleaning and preparing customer data before model prediction	High
3	Lead Scoring Engine	Python, Machine Learning Algorithms	Used for assigning priority scores to potential leads	High
4	Recommendation Module	Python, AI Algorithms	Generates suggestions for prioritizing leads and improving conversion	High
5	Streamlit User Interface	Streamlit, HTML, CSS	Provides user interaction and displays lead analysis results	Medium
6	Dataset Loader & Database Module	Pandas , SQLite/CSV	Used for storing and retrieving customer and lead	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure with separate modules for data processing, prediction, scoring, and user interface.
Readability	5	Clear function names, meaningful variables, and understandable implementation logic.
Reusability	5	Reusable modules for preprocessing, lead prediction, scoring, and recommendations.
Documentation / Comments	4	Good inline comments are provided; additional developer documentation can further improve maintainability.
Overall Score	4.8/5	The project demonstrates excellent coding practices with a modular architecture, reusable components, and clean maintainable code suitable for future improvements.